# Technical Details for Full Proposal

## 1. Origin of the Proposal:

Rotorcraft—including helicopters, multicopters, and VTOL platforms—operate in complex aerodynamic environments involving strong nonlinearities, unsteady flows, and interactions between rotors and the fuselage. These effects are especially critical during low-speed manoeuvres, descent, and operations near the ground. Most existing simulation tools use simplified aerodynamic models, which are inadequate for capturing these regimes, limiting their utility in flight control development, performance prediction, and simulator-based training.

In India, operational demands add further complexity. Indian defence forces routinely fly helicopters and drones at extreme altitudes, often under high-load and low-density conditions. The Ministry of Defence has emphasized the need for heavy-payload logistic drones suited for high-altitude transport—platforms for which no globally available solution currently exists. Meeting such requirements necessitates the development of indigenous aerodynamic modelling tools and simulation frameworks tailored to these environments.

Ganguli (2010) highlighted a persistent gap in India's rotorcraft research, especially in applied aerodynamics and acoustics. Despite advancements in dynamics and control, India still lacks validated aerodynamic tools for helicopter platforms such as ALH, LUH, and LCH. With the rapid deployment of advanced rotorcraft and drone systems, high-fidelity simulators are becoming essential for both control system development and operator training. However, accurate aerodynamic models for critical regimes such as **vortex-ring state (VRS), tail rotor effectiveness, and ground effect** are not yet available.

Tail rotor effectiveness has emerged as a major safety issue. A detailed study by Airbus Helicopters (Dequin, 2019) found that over 18 helicopter accidents per year globally are attributed to unanticipated yaw events, commonly referred to as Loss of Tail Rotor Effectiveness (LTE). An analysis of 310 accidents between 2000–2016 showed that 75% occurred close to the ground, where standard recovery procedures were ineffective. These events are not caused by mechanical failure but by aerodynamic instability and delayed or inadequate pilot response. The study underscores the need for better aerodynamic understanding and real-time modelling of yaw dynamics under crosswind and high-power conditions—elements that are currently missing in most simulation frameworks.

This proposal addresses these challenges by developing mid-fidelity, physics-based aerodynamic models for rotorcraft for the three phenomena, validated through targeted wind tunnel experiments and high-resolution flow diagnostics. These models will be reduced to real-time-compatible forms and integrated into NAL's modular rotorcraft flight dynamics simulator.

**The key objectives are**:

- Develop mid-fidelity models using vortex-particle methods
- Validate them through force/moment and flow field experiments
- Derive real-time reduced-order models for simulator integration

By combining computational and experimental approaches, the project will deliver validated aerodynamic modules tailored for Indian conditions, supporting improved simulator realism, safer control designs, and better pilot preparedness across civilian and defence rotorcraft platforms.

## 2. State of Art Review in the Proposed Subject

### 2.1 Review the status and highlight the significance of the proposed work:

#### 1. Scientific Rationale

Rotorcraft exhibit complex aerodynamic phenomena that are nonlinear, unsteady, and highly interactive, especially in critical flight regimes such as **vortex ring state (VRS)**, **loss of tail rotor effectiveness (LTE)**, and **ground effect transitions**. Conventional modelling approaches, including blade element momentum theory (BEMT) and lifting-line methods, fall short in accurately predicting these effects due to their simplifying assumptions [1, 2]. At the other end, high-fidelity computational fluid dynamics (CFD) models can capture these behaviors but are computationally intensive, making them unsuitable for real-time applications in flight dynamics simulation and control design [4].

The proposed research addresses this fundamental modelling gap by introducing a mid-fidelity, physics-based modelling framework that is both validated experimentally and suitable for real-time integration through reduced-order modelling (ROM). This dual focus—on accuracy and computational efficiency—underpins the scientific rationale. By employing vortex-particle methods validated against wind tunnel experiments using advanced PIV techniques, the proposed work will produce robust aerodynamic models that retain essential nonlinear physics and are amenable to ROM extraction. This approach ensures that dynamic flight interactions—such as induced flow recirculation in VRS or sidewash-driven LTE effects—are accurately represented in simulator environments, control systems, and design workflows.

#### 2. Recent Work in the Area

Internationally, extensive work has been conducted on rotorcraft aerodynamics and simulation. Wakefield and Newman [3] provided foundational insights into helicopter dynamics in the VRS regime, while Johnson [7] introduced analytical models to capture the onset and effects of vortex ring interactions. More recent studies such as Rimša and Liugas [8] used unsteady CFD to examine vortex ring structures in steep descent conditions, and Schmidt et al. [9] highlighted the fluid dynamic analogies between VRS and jet-in-counterflow instabilities. Brown [16] observed in his study that eVTOL aircraft are more susceptible and they show counter-intuitive behaviour to VRS.

In ground effect modelling, Bhagwat and Leishman [5] applied free-vortex wake models to characterize rotor interactions with the surface boundary layer, while Ananthan and Leishman [6] demonstrated the predictive utility of vortex filament-based methods. Matus-Vargas et al. [15] studied ground effect specifically for unmanned aerial vehicles and highlighted the need for accurate modelling of the same.

Similarly, in the context of tail rotor effectiveness, Saleh et al. [10] and Cuzieux et al. [11] presented physics-based and data-driven insights into LTE-related flight anomalies, showing the need for more predictive modelling.

Recent advances in CFD frameworks such as SU2 have enabled more accessible rotorcraft simulations [12], while comprehensive simulation tools have been developed to model full rotorcraft flight dynamics [13]. However, Spalart and Venkatakrishnan [4] emphasized the ongoing challenges in translating CFD insights into real-time tools due to resolution demands and turbulence modelling uncertainties.

In the Indian context, research led by Ganguli [14] has outlined rotorcraft dynamics and design efforts, yet the literature lacks robust efforts focused on real-time aerodynamic modelling or ROMs for flight dynamics. Most Indian rotorcraft studies have emphasized structural or control aspects, with limited systematic progress on aerodynamics-informed flight simulation or high-resolution flow experiments.

### 3. Gaps in Current Research

Despite notable advancements in rotorcraft aerodynamics, several critical gaps persist in current research. A primary limitation is the absence of mid-fidelity aerodynamic models specifically optimized for reduced-order model (ROM) extraction. While high-fidelity simulations and comprehensive CFD tools exist, they are rarely designed with the computational and structural constraints necessary for ROM generation, thereby limiting their deployment in real-time applications such as flight simulation and control design [4, 7, 12]. Moreover, a substantial portion of existing CFD-based studies—particularly those addressing complex phenomena such as vortex ring state (VRS) and loss of tail rotor effectiveness (LTE)—lack rigorous experimental validation, diminishing their predictive reliability for safety-critical scenarios [3, 4, 8].

In addition, there remains a dearth of real-time capable aerodynamic models that can accurately capture the nonlinear, unsteady flow dynamics inherent in rotorcraft operations. Current ROMs are often overly simplified or insufficiently robust to generalize across the wide range of operational regimes, including transitions into and out of VRS, tail rotor failures, and ground effect conditions [7, 10, 11]. This oversimplification undermines their utility in high-fidelity flight training and performance prediction environments. Furthermore, much of the existing research exhibits a fragmented methodology, wherein computational and experimental approaches are pursued in isolation. The lack of an integrated simulation–experiment–ROM workflow reduces the reliability and transferability of the resulting models.

Within the Indian context, the gap is even more pronounced. As outlined by Ganguli [14], rotorcraft research in India has traditionally focused on structural dynamics and control design, with limited emphasis on aerodynamic modelling frameworks that combine simulation and experimental data for ROM development. Given India's strategic push for indigenous aerospace capabilities and its increasing reliance on rotorcraft platforms in both defence and civil sectors, the lack of robust, validated, and real-time aerodynamic models represents a significant technological shortcoming. Addressing these interconnected gaps is essential for advancing rotorcraft simulation fidelity, enhancing pilot training, and ensuring the safe operation of next-generation vertical lift platforms.

## 2.2 Importance of the proposed project in terms of impact and innovation

The proposed project addresses a critical gap in rotorcraft aerodynamic modeling by developing validated, mid-fidelity, physics-based aerodynamic models and reduced-order models (ROMs) specifically designed for real-time integration into rotorcraft flight dynamics frameworks. The innovation lies in bridging the divide between high-fidelity CFD simulations—which are accurate but computationally expensive—and low-fidelity models that fail to capture nonlinear aerodynamic phenomena like vortex-ring state (VRS), tail rotor effectiveness (LTE), and ground effect.

This project uniquely integrates vortex-particle-based simulation with experimental validation through wind tunnel testing and high-resolution Particle Image Velocimetry (PIV). This combined

approach allows for the systematic derivation of ROMs that are both physically grounded and computationally efficient, enabling their direct use in simulators, control design, and operator training. Such a comprehensive modeling-to-simulation pipeline does not currently exist in India, and few examples exist globally that link medium-fidelity modeling with real-time-capable ROM development in this domain.

The project will support and directly feed into the ongoing rotorcraft simulation and control modelling efforts at NAL, including the development of the "High Fidelity Rotorcraft Flight Dynamic Modelling and Control Framework" and the recently introduced SimCopter aeromechanics toolbox. The proposed models will provide the aerodynamic backbone for these initiatives, enabling more accurate prediction of handling qualities, control responses, and training scenarios.

Further, this work sets the stage for future commercialization through a potential startup incubated at NAL or IIT Bombay, targeting the development of logistics drones for high-altitude operations—a need already highlighted by the Ministry of Defence and for which no global solution currently exists. The aerodynamic models developed in this project will be essential to the performance prediction, control design, and virtual testing of such platforms.

By significantly improving the aerodynamic fidelity of real-time flight dynamics simulations, this project will enhance safety, training effectiveness, and design reliability. The innovation will benefit simulator developers, flight control engineers, rotorcraft designers, and training institutions. Within the Indian context, where new helicopter platforms are being inducted and air mobility applications are growing, the impact extends to national self-reliance, aerospace innovation, and operational preparedness.

In summary, the project not only advances the scientific state-of-the-art but also offers high-impact deliverables for national aerospace programs, emerging drone technologies, and the growing ecosystem of rotorcraft applications in both civil and defence sectors.

### 2.3 Proof of concept can be shared with appropriate preliminary data

Preliminary research conducted by the group on modelling aerodynamic interactions in coaxial rotor helicopters in ground effect serves as a strong proof of concept for the proposed project and demonstrates readiness to extend these methods to broader rotorcraft configurations and critical flight regimes. In a recent study, aerodynamic interactions between coaxial rotors and the fuselage under ground effect were investigated using two medium-fidelity approaches: the actuator disk method in ANSYS Fluent and the vortex particle method via the DUST solver. The models were validated against experimental data for a coaxial rotor–fuselage configuration and later extended to a more realistic setup involving a four-bladed rotor and the ROBIN helicopter fuselage.

The focus of the study was primarily on hovering conditions in ground effect. The methodologies and modelling frameworks developed through this research also offer a good foundation for future work involving forward flight and more complex aerodynamic regimes such as vortex-ring state or lateral inflow during tail rotor operations.

The following figure presents selected results showing velocity magnitude contours from the study on ground effect using actuator disk model technique in Ansys and vortex particle method in DUST.

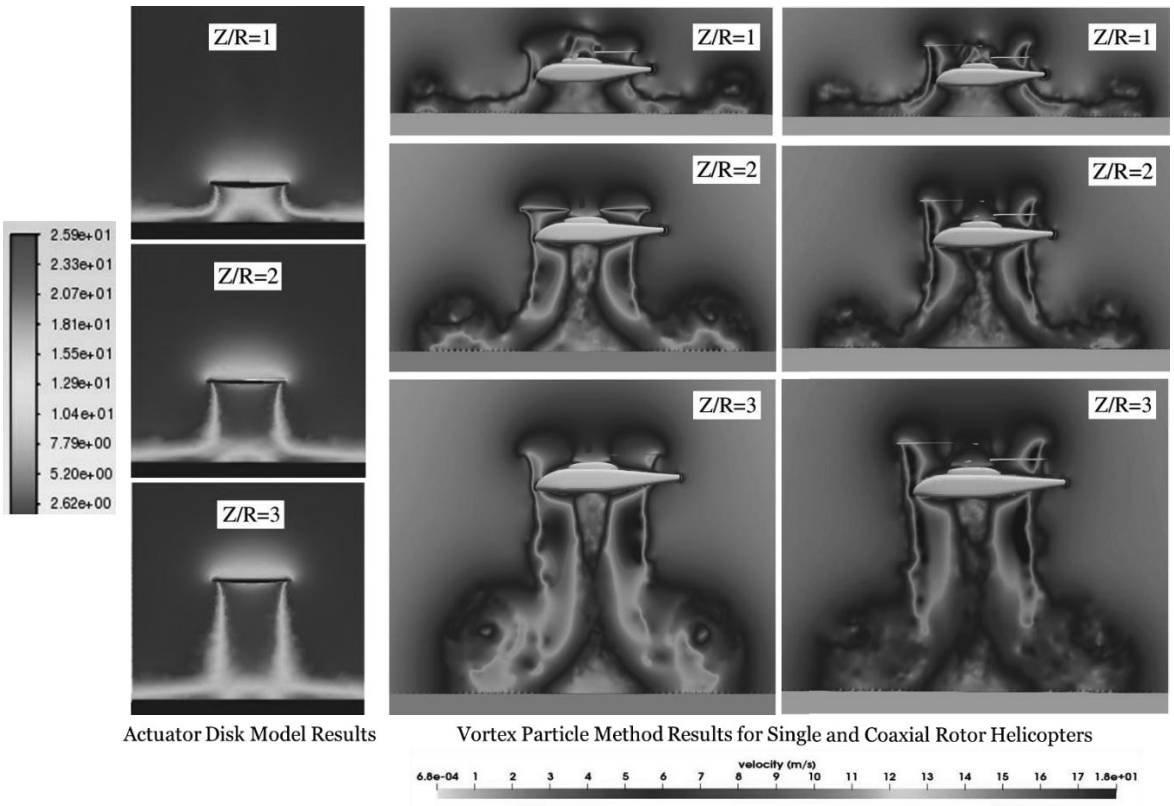



***Figure 1**. Simulation Results from the Research on Coaxial Rotor Aerodynamics in Ground Effect.*

In parallel, initial work has also been initiated in collaboration with NAL on the development of SimCopter, a modular, GUI-based flight dynamics and control simulation framework designed to support real-time rotorcraft modeling. SimCopter currently includes low-fidelity aerodynamic models, blade element-based rotor modules, and stability and handling qualities analysis using ADS-33E-PRF metrics. It supports both conventional helicopters and multirotor configurations. The long-term roadmap includes integrating mid-fidelity aerodynamic models and reduced-order modules for real-time simulation and control prototyping.

**2.4 If the project is location specific, basis for selection of location be highlighted:**

The project is not location specific.

## 3. Work Plan:

**3.1 Methodology**:

**a) Literature Survey**

The project will begin with a thorough literature review to build a foundational understanding of the aerodynamic phenomena and modelling techniques relevant to rotorcraft flight. Special focus will be given to studies on vortex-ring state (VRS), tail rotor effectiveness (LTE), and ground effect interactions. This review will encompass both classical models such as blade element momentum theory (BEMT) and recent developments in computational and experimental research. Literature on vortex-particle methods and wake dynamics will be studied in detail. The goal is to identify validated physics and modelling approaches that can be reused or extended, while also pinpointing gaps that the current project can address through new experiments and simulations. Insights from

this review will directly inform the design of simulation cases, experimental test setups, and the structure of the reduced-order models (ROMs).

**b) Theoretical Modelling**

As a theoretical foundation, a custom implementation of blade element momentum theory (BEMT) will be developed. This analytical tool will serve both as a benchmark for mid-fidelity simulation results and as a base framework on which the aerodynamic models will be extended. Enhancements to BEMT, including non-uniform inflow, dynamic inflow lag, and unsteady aerodynamic effects, will be incorporated to ensure relevance to real rotorcraft flight conditions. The modular nature of the implementation will also allow easy integration of new empirical corrections or physics-based enhancements derived from experiments and simulations.

**c) Experiments**

An instrumented test rig will be developed, consisting of multiple rotors with feedback loop control to various flight conditions of rotorcraft. The test rig will utilize a 6-degree-of-freedom force and moment balance sensor to measure thrust, torque, and aerodynamic moments.

An adjustable ground plane will help in studying the performance of rotors in ground effect. The wind tunnel facilities at IIT Bombay will be utilized to create vertical descent like conditions to trigger vortex-ring-state. The wind tunnel facilities will also be used to generate yawed flight / lateral translation like conditions to test and model tail rotor effectiveness for a conventional helicopter type configuration.

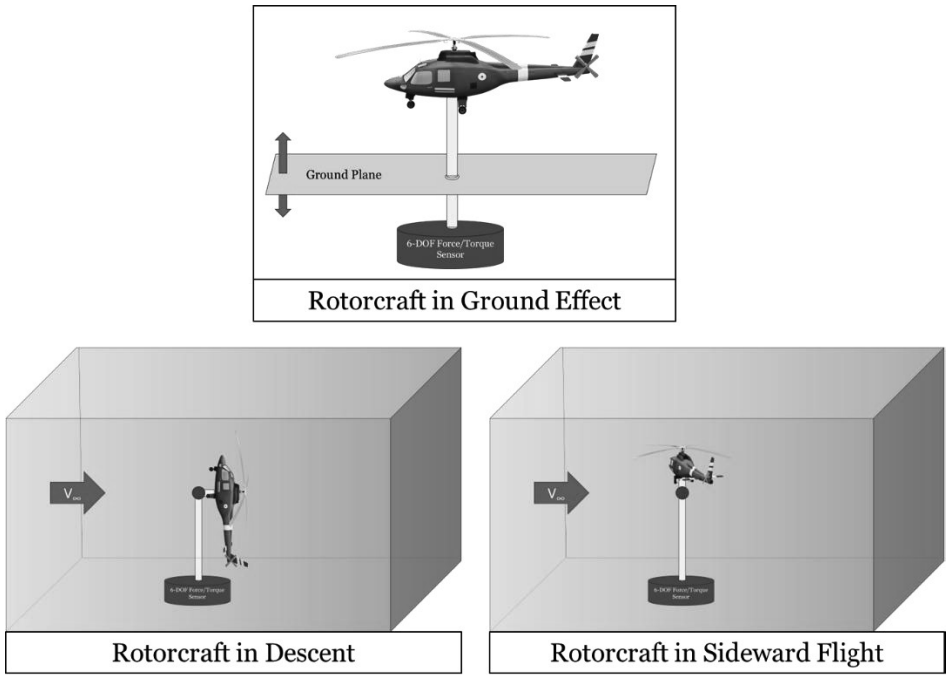



***Figure 2**. Schematic representation of experimental study of Ground effect, Vortex ring state, and Tail rotor effectiveness in lateral flight*

These experiments will provide critical data on rotor performance under controlled and repeatable laboratory conditions.

In parallel, flow diagnostics will be conducted using high-resolution Particle Image Velocimetry (PIV) available in the lab. This will allow direct visualization of tip vortex trajectories, wake formation, and ground effect interactions. The use of PIV will provide both qualitative and quantitative insights into the flow physics responsible for the measured aerodynamic forces and moments. These results will guide model development, help interpret simulation outputs, and serve as a validation dataset.

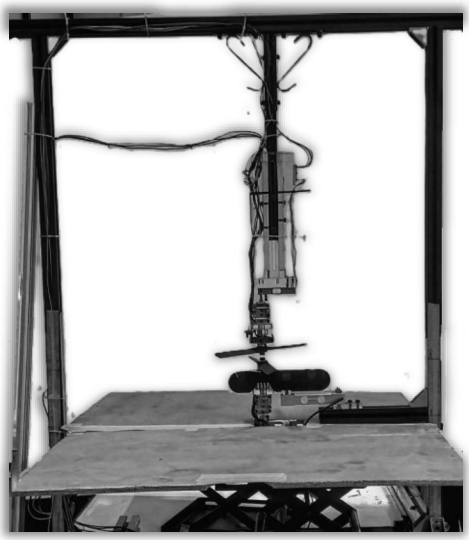


***Figure 3.*** *Experimental setup for research on coaxial rotor helicopter in ground effect.*

**d) Simulations**

To complement experimental work, mid-fidelity aerodynamic simulations employing Vortex Particle Methods (VPM) will be carried out. There are two open source VPM codes available at present. One is DUST, an open-source vortex-particle method developed by the Department of Aerospace Science and Technology at Politecnico di Milano. Another one is FLOWUnsteady developed by Brigham Young University. Tools like these are well-suited for simulating unsteady rotor wakes and aerodynamic interactions at a manageable computational cost. Simulations will be set up to match experimental conditions and then extended to cover a broader range of inputs that are impractical to test physically. Simulation outputs will include unsteady inflow distributions, time-resolved blade-element forces and moments, and wake structures.

Simulation results will be validated against both the experimental data obtained in this project and benchmark cases from the literature. This dual-source validation strategy will ensure that the simulation framework is reliable and can be confidently used to generate data for reduced-order model development. Particular emphasis will be placed on generating datasets for flight regimes like VRS and LTE, where high-fidelity CFD is impractical, and low-fidelity models are inaccurate

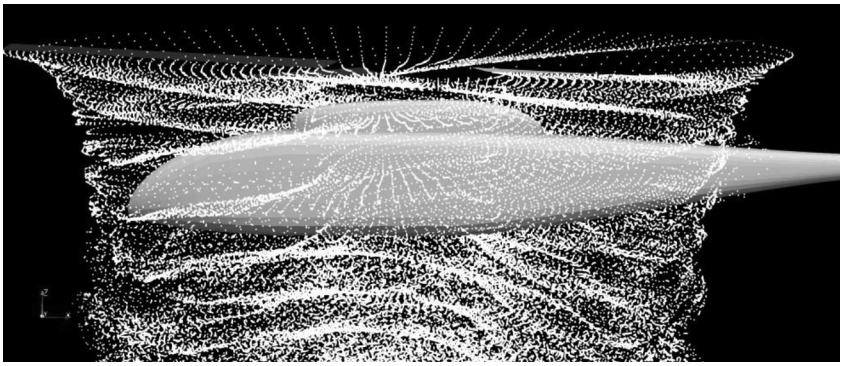


***Figure 4**. Simulation result of a single rotor helicopter using Vortex Particle Method tool.*

### e) Reduced-Order Model Development

The next phase of the project will focus on developing reduced-order models (ROMs) from the validated experimental and simulation datasets. The ROMs will aim to capture critical aerodynamic behaviours such as non-uniform and unsteady inflow patterns and blade-section forces and moments. These models will be formulated using a combination of physics-based analytical expressions and data-driven techniques, ensuring both accuracy and computational efficiency.

Initial model development will rely on insights from theory and validated simulation data to maintain physical interpretability. For highly nonlinear phenomena or data-rich regions, machine learning techniques such as polynomial regression, neural networks, or surrogate modelling may be explored, while ensuring generalizability across flight conditions. Proper Orthogonal Decomposition (POD) and other model reduction techniques will be used to simplify high-dimensional simulation data into compact, real-time-capable models suitable for integration with flight dynamics simulators. The final ROMs will output aerodynamic loads as functions of control inputs, inflow conditions, and rotor states, enabling use in dynamic simulation and control design environments.

### f) Integration with Flight Dynamics Simulators

Throughout the project, development will proceed in close coordination with NAL to ensure that the aerodynamic models and ROMs are tailored for integration into their flight dynamics simulation framework. Regular technical review meetings and collaborative testing will ensure compatibility, usability, and iterative refinement. Final model implementations will be delivered in well-documented, modular formats, with support provided for integration, validation, and future extension. Select findings, simulation animations, and flow diagnostics will also be disseminated through academic publications, conferences, and joint seminars with NAL to promote broader impact and industry adoption.

In the event of unforeseen delays in experimental work, the project will leverage validated rotorcraft datasets from the literature (e.g., Caradonna–Tung rotor, NASA Tiltrotor Test Rig) to perform surrogate validation.

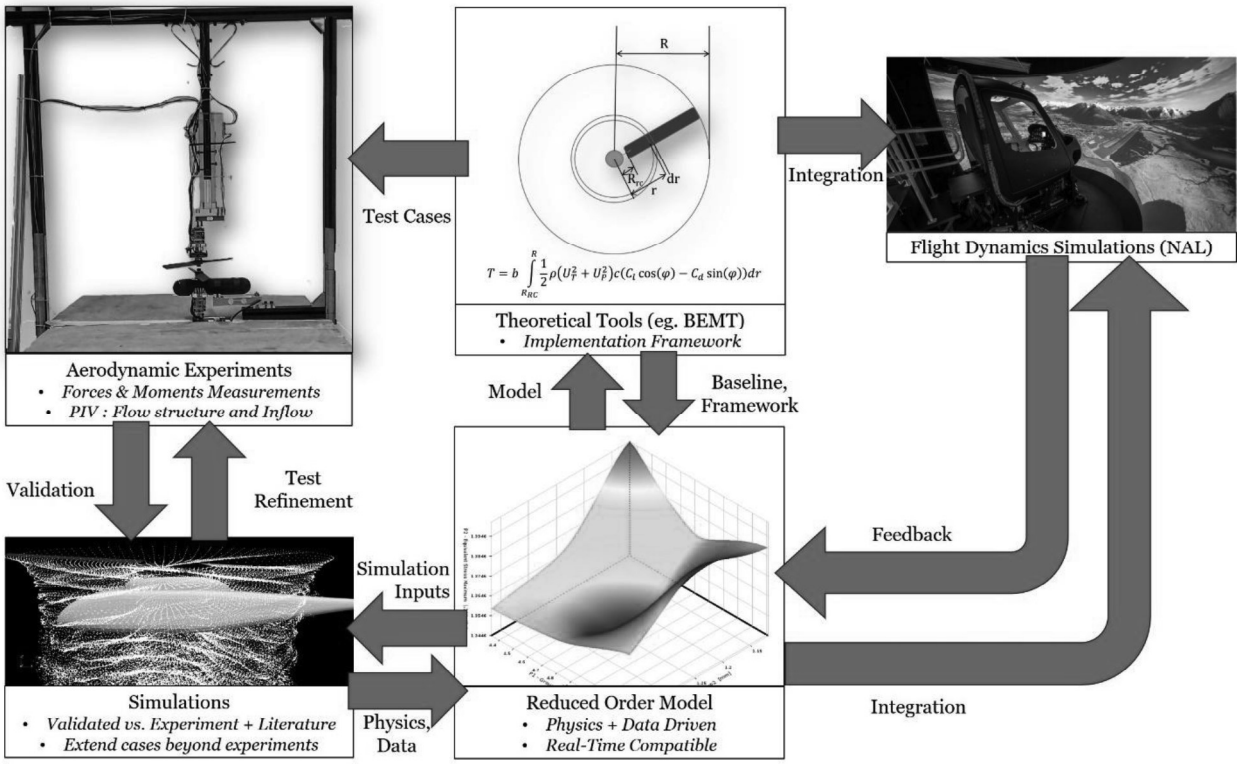



***Figure 5**. Overall methodology flowchart*

## 3.2 Suggested Plan of action for utilization of research outcome for social economic growth benefits expected from the project.

The proposed research outcomes—validated mid-fidelity aerodynamic models and real-time-capable reduced-order models (ROMs)—will contribute significantly to India's aerospace and defence sectors by enhancing rotorcraft safety, reducing design and development risks, and fostering indigenous innovation. These models will be directly integrated into the modular rotorcraft flight dynamics simulation framework under development at the National Aerospace Laboratories (NAL), in close collaboration with their Flight Mechanics and Control Division. This ensures immediate and practical deployment of the research outputs in simulation-based control design, handling qualities assessment, and pilot training.

Importantly, the present project complements and integrates with a parallel initiative titled "High Fidelity Rotorcraft Flight Dynamic Modelling and Control Framework" at NAL, led by Dr. Omkar Halbe (Honorary co-PI of the present proposal). The proposed aerodynamic models will serve as core components of this broader framework, enabling high-fidelity, real-time simulation tools tailored for Indian rotorcraft platforms.

By reducing dependence on imported modelling tools and creating validated, open-access methodologies, the project will support India's strategic goal of technological self-reliance. In the longer term, the outcomes will enable safer and more efficient operations across sectors such as defence aviation, emergency medical services, disaster relief, and urban air mobility. This, in turn, will drive job creation in aerospace software development, support small and medium

enterprises (SMEs) in simulation and training systems, and enhance national preparedness in both civil and defence domains—contributing meaningfully to India's broader socioeconomic growth.

### 3.3 Environmental impact assessment and risk analysis.

The proposed project involves computational modelling and controlled wind tunnel experiments; thus, direct environmental impacts are minimal and confined to laboratory settings. Experimental tests utilize established facilities and standard measurement techniques (e.g., Particle Image Velocimetry), posing no significant environmental hazards. Indirectly, however, the project will positively impact the environment by enabling accurate aerodynamic models for rotorcraft, contributing to improved fuel efficiency, reduced emissions, and optimized flight operations, thereby aligning with sustainable aviation goals.

Risk analysis primarily pertains to technical and implementation risks, including potential delays in validating models or unforeseen complexities in aerodynamic phenomena. To mitigate these risks, the project incorporates phased deliverables, periodic review mechanisms, and collaborative validation efforts with NAL. Furthermore, the project's methodological reliance on open-source software and existing experimental facilities significantly reduces the probability of unforeseen resource constraints. Overall, environmental and operational risks remain low, manageable, and adequately addressed through rigorous planning and structured monitoring.

### 3.4 Dissemination plan :-

The results of the project will be disseminated through a multi-pronged strategy aimed at reaching both the scientific community and relevant stakeholders in the aerospace industry and defence sectors. Technical outcomes—such as aerodynamic models, validation data, and reduced-order modelling frameworks—will be published in reputed peer-reviewed journals (e.g., Journal of the American Helicopter Society, Aerospace Science and Technology, Physics of Fluids) and presented at international conferences such as Vertical Flight Society and AIAA Aviation.

The other important focus of the proposed project is the effective transfer of aerodynamic models and reduced-order tools to support the development of a real-time rotorcraft flight dynamics simulator at NAL. The validated mid-fidelity aerodynamic models and real-time-capable ROMs will be delivered in modular, well-documented formats directly integrable into NAL's simulator framework. Joint technical review meetings, code walkthroughs, and integration workshops will be conducted to ensure smooth technology transfer and adoption by NAL's development team.

### 3.5 Time Schedule of activities giving milestones through BAR diagram

| | **Timeline Months** | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Activity** | **3** | **6** | **9** | **12** | **15** | **18** | **21** | **24** | **27** | **30** | **33** | **36** |
| Project staff hiring | | | | | | | | | | | | |
| Equipment procurement | | | | | | | | | | | | |
| Simulation setup establishment | | | | | | | | | | | | |
| Experimental setup development | | | | | | | | | | | | |
| Force and moment measurements | | | | | | | | | | | | |
| Flow measurements | | | | | | | | | | | | |
| Simulations | | | | | | | | | | | | |

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|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Validation and data interpretation | | | | | | | | | | | | |
| ROM development and validation | | | | | | | | | | | | |
| Dissemination and handover | | | | | | | | | | | | |

## 3.6 Bibliography

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